www.tntextbooks.in

Sun. The Sun to Earth distance is appriximately 150 million km. (Assume the orbit of Earth to be circular)

The centripetal acceleration $a_c = \frac{v^2}{r}$

V - velocity of Earth around the orbit

r - radius of orbit or distance of Earth to Sun

Velocity of Earth is written in terms of angular velocity (ω) as

$$v = \omega r$$

By substituting in the centripetal acceleration formula, $a_c = \frac{\omega^2 r^2}{r} = \omega^2 r$

But $\omega = \frac{2\pi}{T}$ where T is time for the Earth to orbit around the sun, which is one year.

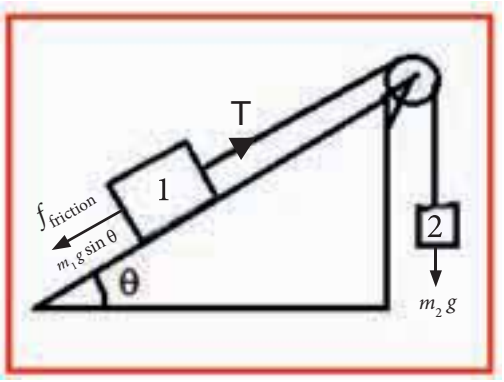
$$\text{T} = 365 \text{ days} = 365 \times 24 \times 60 \times 60 \text{ s} = 3.1 \times 10^7 \, s$$

$$\omega = 2.02 \times 10^{-7} \, rad \; per \; sec$$

$$a_c = \left(2.02 \times 10^{-7}\right)^2 \times \left(150 \times 10^8\right)$$

$$a_c = 6.12 \times 10^{-4} \, ms^{-2}$$

11. A block 1 of mass m_1, constrained to move along a plane inclined at angle θ to the horizontal, is connected via a massless inextensible string that passes over a massless pulley, to a second block 2 of mass m_2. Assume the coefficient of static friction between the block and the inclined plane is μ_s and the coefficient of kinetic friction is μ_k



What is the relation between the masses of block 1 and block 2 such that the system just starts to slip?

Solution:

For all parts of this problem, it will be convenient to use different coordinate systems for the two different blocks. For block 1, take the positive x -direction to be up the incline, parallel to the plane, and the positive y -direction to be perpendicular to the plane, directed with a positive upward component. Take the positive direction of the position of block 2 to be downward.

The normal component N of the contact force between block 1 and the ramp will be

$$N = m_1 g \cos\theta \, . \qquad (1)$$

The net x -component of the force on block 1 is then

$$F_{1x} = T - f_{\text{friction}} - m_1 g \sin\theta \qquad (2)$$

where T is the tension in the string

For the just-slipping condition, the frictional force has magnitude

$$f_{\text{friction}} = \mu_s N = \mu_s m_1 g \cos\theta. \qquad (3)$$

The tension in the string is the gravitational force of the suspended mass,